

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc BIOCHEMISTRY

SEMESTER III EXAMINATION (BACKLOG)

BC804: ADVANCED RECOMBINANT DNA TECHNOLOGY AND BIOPHARMACEUTICALS

Date: 25.04.2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1. Choose the correct option and put ✓ mark in front of it

1. Which of the following techniques uses digestion with a restriction endonuclease?
 - A. Southern blotting
 - B. Polymerase chain reaction
 - C. DNA Sequencing
 - D. Northern blotting
2. Which of the following methods cannot be used to introduce a specific mutation at a predetermined site in a DNA sequence?
 - A. Primer extension on a single stranded template using a primer that incorporates a base mismatch
 - B. Primer extension on a single stranded template using an error-prone DNA polymerase
 - C. PCR amplification using a double stranded template with one primer incorporating a base mismatch
 - D. None of the above
3. Positional cloning refers to
 - A. Using a selection procedure to clone a cDNA
 - B. Cloning a portion of a gene using PCR
 - C. Isolating a gene by PCR using primers from another species
 - D. Mapping a gene to a chromosomal region, identifying and cloning
4. STR (Simple tandem repeat) polymorphisms in humans are mainly used to
 - A. Transfer disease resistance factors
 - B. Estimate blood transfusions matches
 - C. Solve paternity and criminal cases
 - D. None of the above
5. Tendency for genes not to be randomized is called -----
 - A. Haplotypes
 - B. Linkage disequilibrium
 - C. Single nucleotide polymorphisms
 - D. None of the above
6. ----- are a small stretch of DNA that are unique in the genome
 - A. STS (Sequence tagged sites)
 - B. Polymorphisms
 - C. Tandem clusters
 - D. None of the above
7. dsRNAs are cleaved into 21-23 nucleotide segments (siRNAs) by the enzyme called
 - A. Polymerase
 - B. Dicer
 - C. Helicase
 - D. None of the above

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8. Antimalarial compound quinine is isolated from the bark of -----
A. Mangifera indica
B. Cinchona officinalis
C. Digitalis purpurea
D. None of the above
9. ----- deals with the interaction of drugs with cellular proteins such as receptors or enzymes
A. Pharmacokinetics
B. Toxicology
C. Pharmacodynamics
D. None of the above
10. ----- is the dose of a drug, which is therapeutically effective in 50% of animals
A. ED₅₀
B. LD₅₀
C. Maximum tolerated dose
D. None of the above
11. The enzymes involved in Phase I reaction of drug metabolism are
A. Cytochrome P450
B. Esterases
C. Amidases
D. All the above
12. ----- is the movement of drugs from its site of administration into the circulation
A. Absorption
B. Distribution
C. Metabolism
D. Elimination
13. ----- the fraction of the administered dose of a drug that reaches the systemic circulation
A. Volume of Distribution
B. Distribution
C. Bioavailability
D. None of the above
14. In vivo absorption of a drug across the gut wall can be obtained by measuring the rate of transport of the drug across the ----- cell line
A. SK Mel 28
B. WM 35
C. Caco-2
D. All the above
15. ----- is the study of variability in drug response determined by multiple genes within the genome
A. Pharmacology
B. Pharmacogenetics
C. Pharmacogenomics
D. Pharmaceutics

Q2. State True or False:

16. Cassette mutagenesis is the replacement a region of DNA with new sequences (True/ False)
17. Biocompatibility is the ability of a material to perform with an appropriate host response in a specific application (True/ False)
18. Genetic factors do not have any role in individual differences of drug metabolism (True/ False)
19. Intellectual property is the legal rights granted by the government for a limited period in exchange of the public disclosure of the inventions (True/ False)
20. Drugs which are lipid insoluble will have a high Vd (volume of distribution) (True/ False)

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Part B

Total Marks: 50

Section I

Q.1 (A) What are the differences between Sanger and Maxam Gilbert methods of DNA sequencing (4)

Q.1 (B) Discuss RNA interference (4)

OR

Q.1 (B) Explain chromosome jumping with the help of diagrams

Q.1 (C) Explain the principle of DNA fingerprinting (2)

Q.2 (A) Write a note on non-coding DNA in eukaryotes (4)

OR

Q.2 (A) Explain the advantages and disadvantages of baculovirus system for the recombinant protein production

Q.2 (B) Explain the role of anchored primers in differential display (4)

OR

Q.2 (B) Explain the principle of Northern blotting

Q.2 (C) Explain **any one** of the following (2)

- i) Gene knockout
- ii) Transgenics

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Section II

- Q.3(A) Explain different types of DNA polymorphisms (4)
- Q.3(B) Discuss the Phase I drug metabolic reactions (4)
- OR**
- Q.3(B) Discuss the preclinical genotoxicity studies
- Q.3(C) Write a short note on drug receptor interaction (2)
- Q.4(A) Explain the plasma stability assay for a drug (4)
- OR**
- Q.4(A) Explain CYP450 inhibition tests and its importance
- Q.4(B) What is the importance of testing the microsomal stability of drug? (2)
- Q.4(C) Explain the kinetics of elimination of a drug from the body (4)
- Q.5(A) Write a note on protein drug delivery system (4)
- OR**
- Q.5(A) Discuss the Hydrogel based drug delivery
- Q.5(B) Discuss the structure of a liposomes (4)
- Q.5(C) Write a short note on DNA vaccines (2)